

ClipMind AI

AI Video Summarization & Key Moments Detection Platform

Project Documentation

A technical and functional overview of the ClipMind AI project, prepared for repository documentation and project submission.

Repository Branch: Inchara-Y

Technology: Next.js / React, FastAPI, Python, SQLite/PostgreSQL-ready configuration

Document Version: 1.0

1. Project Overview

ClipMind AI is an AI-powered video summarization and key-moment detection platform designed to help users process educational and informational videos efficiently. The platform supports video upload, processing, transcript generation, AI-assisted summarization, keyword extraction, key-moment identification, bookmarking, analytics, and role-based access.

2. Objectives

The project aims to reduce the time required to consume long-form video content by presenting concise summaries, searchable transcripts, important moments, and relevant keywords. It also provides separate experiences for Creators, Learners, Educators, and Administrators.

3. User Roles

Creator: uploads and manages video content and can view generated summaries, transcripts, key moments, and analytics.

Learner: browses available content, watches videos, reads summaries and transcripts, and uses bookmarks/history features.

Educator: manages educational content and uses summarized material and analytics to support learning activities.

Administrator: manages users, monitors platform activity, and accesses administrative analytics.

4. Core Features

- User registration and login
- Role-based access control
- Video upload and management
- Video processing workflow
- Transcript generation
- AI-generated summaries
- Keyword extraction
- Key-moment detection
- Video watch interface
- Bookmarks and learning history
- Creator/learner dashboards
- Administrator dashboard and analytics
- About page
- Responsive frontend interface
- REST API backend
- SQLite development database with PostgreSQL-ready configuration
- Docker configuration for deployment

5. System Architecture

The application follows a frontend-backend architecture. The frontend is implemented using Next.js/React and provides authentication screens, dashboards, video pages, upload functionality, analytics, and the About page. The backend is implemented using FastAPI and exposes REST endpoints for authentication, users, videos, transcripts, summaries, and related functionality. The backend also contains the AI service layer used by the video-processing workflow.

6. Technology Stack

Frontend: Next.js, React, JavaScript/JSX, CSS.

Backend: Python, FastAPI.

Data layer: SQLite for straightforward local development; the project also contains PostgreSQL/Docker-oriented configuration.

AI/processing layer: Python-based AI service abstraction for transcription, summarization, keyword extraction, and key-moment processing.

Development/deployment: npm, Python virtual environment, Uvicorn, Docker Compose.

7. Frontend Structure

The frontend contains route-based pages for the main application areas. Major pages include the home page, login, registration, dashboard, upload, watch, and About page. The dashboard experience is adapted to the user's role and provides navigation to the relevant features.

8. Backend Structure

The backend contains the FastAPI application entry point, database configuration, data models, authentication utilities, and an AI service module. API routes are responsible for user authentication and application operations such as video handling and generated content.

9. Authentication and Security

The project includes user registration and login with password protection and role information. Password handling is implemented without relying on the bcrypt/passlib compatibility issue encountered during development.

Environment-specific secrets should be stored in environment files and must not be committed to the public repository.

10. Video Processing Workflow

A typical workflow is: (1) user uploads a video, (2) the backend accepts and processes the video, (3) transcript information is generated, (4) summary and keywords are produced, (5) important/key moments are identified, and (6) the frontend presents the resulting information to the user.

11. AI Features

The application includes an AI service abstraction for transcript generation, summarization, keyword extraction, and key-moment detection. The repository is structured so that the AI processing layer can be connected to production models/services as required. The current project package is intended to provide a runnable platform and a replaceable AI-service layer.

12. Installation and Local Setup

Backend:

1. Open a terminal in the backend directory.
2. Create a virtual environment: `python -m venv venv`
3. Activate it on Windows: `.\venv\Scripts\Activate`
4. Install dependencies: `pip install -r requirements.txt`
5. Start the API: `uvicorn main:app --reload`

Frontend:

1. Open a second terminal in the frontend directory.
2. Install dependencies: `npm install`
3. Start the development server: `npm run dev`
4. Open the local Next.js address shown by the terminal.

13. Project Repository Guidelines

The repository should contain source code, documentation, configuration examples, and project assets that are safe to publish. Do not commit virtual environments, `node_modules`, build output, local databases, uploaded user videos, or secret environment variables. The included `.gitignore` should be used and reviewed before the first Git commit.

14. Suggested GitHub Repository Contents

Recommended repository structure:

ClipMind-AI/

- backend/
- frontend/
- docker-compose.yml
- README.md
- .gitignore
- project documentation/

The documentation PDF can be stored in a `documentation` or `docs` folder so reviewers can easily understand the system before examining the source code.

15. Future Enhancements

The platform can be extended with production-grade Whisper transcription, transformer-based summarization, stronger video key-moment algorithms, cloud object storage, production PostgreSQL deployment, background job queues, richer analytics, automated testing, CI/CD, and improved security controls.

16. Conclusion

ClipMind AI provides an end-to-end foundation for an AI-assisted video learning and content-processing platform. Its modular frontend, FastAPI backend, role-based access model, video workflow, and replaceable AI service layer make it suitable for continued development and deployment.

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